

RITESH PANDEY

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✉ Email

🌐 Linkdin

🐙 Github

👤 GeeksForGeeks

📄 Leetcode

👤 Coding Ninja

PROFILE SUMMARY

Pre-final year Computer Science student with strong foundations in Data Structures, data analysis, and Machine Learning. Experienced in building data-driven applications and ML-based systems, with hands-on experience in data preprocessing, visualisation, and model evaluation. Interested in AI systems, data analytics, and real-world deployment of data-driven solutions.

EDUCATION

Indian Institute of Information Technology, Kottayam

2023 – 2027

B.Tech CSE with Specialization in Cyber Security

Kottayam, Kerala

Relevant Coursework :

Data Science and Data Analytics, ML and DL, Cloud Computing, DBMS, Computer Networks, Cryptography

PROJECTS

ChatInsight 🗒 | T5, PyTorch, HuggingFace Transformers, NLP, Seq2Seq, Deep Learning. **Sept - Oct 2025**

- Built an end-to-end data preprocessing pipeline (cleaning, tokenization, batching) using Pandas and Regex.
- Designed evaluation workflow using ROUGE-based metrics to assess model performance.
- Trained model using HuggingFace Trainer API with AdamW optimizer and weight decay tuning.
- Built an end-to-end data preprocessing pipeline (cleaning, tokenization, batching) using Pandas and Regex.
- Developed an abstractive summarization system using T5 Transformer for multi-turn dialogue summarization.

SuggestBot 🗒 | Streamlit, SQLite, LangChain, Ollama(LLaMA2), Gemini API, RapidFuzz **Jan - Apr 2026**

- Built Natural Language-to-SQL pipeline achieving 92% structured query accuracy on healthcare dataset.
- Developed a context-aware chatbot with Gemini API and SQLite-based chat history multi-turn interactions.
- Reduced query failure rate by 40% using typo-tolerant fuzzy matching (RapidFuzz).
- Integrated FAISS-based vector retrieval enabling contextual multi-turn reasoning.
- Designed an end-to-end NL → SQL → DB pipeline for real-time query execution and response generation.

GreenShift 🗒 | Chart.js, FastAPI, Uvicorn, NumPy, Pandas, MDP

Feb - Mar 2026

- Developed an MDP-driven system to align workload scheduling with renewable energy availability.
- Applied value iteration to derive optimal VM allocation strategies under changing demand.
- Built FastAPI services to simulate system behavior and evaluate optimization policies.
- Processed and integrated real-world workload datasets with energy consumption models.
- Designed comparative analysis across scheduling strategies using key energy efficiency metrics.
- Presented insights through interactive visualizations of performance and convergence behavior.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript

Frontend: React.js, TypeScript, Tailwind CSS, Framer Motion

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Databases: PostgreSQL, MySQL, MongoDB, Supabase

Machine Learning: Linear and Logistic Regression, PyTorch, HuggingFace, Transformers, T5, Scikit-learn

Data Science: Pandas, NumPy, Statistics and Probability, Data Wrangling and Preprocessing, Data

Visualisation, BeautifulSoup and Scrapy

Cloud Tools: AWS (EC2, S3, IAM, Lambda, RDS, VPC, CloudWatch), Postman, Git/GitHub, Linux, Figma

ACHIEVEMENT

- Solved 600+ DSA problems on platforms like Coding Ninja(114), LeetCode(229), and GFG(293).
- Secured Rank: 16 on GeeksforGeeks among students of my college.
- Earned IBM certification in Basics of Quantum Information.
- Completed 10-hour Cybersecurity training via Skill India (Tech Mahindra Foundation).
- Participated in a 5-day FDP on RNLP at IIIT Kottayam, actively engaging in discussions and problem-solving.
- Supervised Machine Learning: Regression and Classification – DeepLearning.AI & Stanford University (Coursera)
- (Lead:Nov 2025 - Present, SubLead:Oct 2024 - Nov 2025) PR Outreach for BetaLabs (Technical Club of IIIT Kottayam),coordinating with students,Sponsors, and stakeholders event participation and ensure smooth execution.
- (SubLead:Sept 2024 - Nov 2025) PR Event Management for WildBeats (Cultural Club of IIIT Kottayam). Managed event promotions, logistics, and student engagement, and improving overall event coordination.